# Galactose-α-1,3-galactose and Delayed Anaphylaxis, Angioedema, and Urticaria in Children

BACKGROUND AND OBJECTIVE: Despite a thorough history and comprehensive testing, many children who present with recurrent symptoms consistent with allergic reactions elude diagnosis. Recent research has identified a novel cause for “idiopathic” allergic reactions; immunoglobulin E (IgE) antibody specific for the carbohydrate galactose-α-1,3-galactose (α-Gal) has been associated with delayed urticaria and anaphylaxis that occurs 3 to 6 hours after eating beef, pork, or lamb. We sought to determine whether IgE antibody to α-Gal was present in sera of pediatric patients who reported idiopathic anaphylaxis or urticaria. METHODS: Patients aged 4 to 17 were enrolled in an institutional review board–approved protocol at the University of Virginia and private practice allergy offices in Lynchburg, VA. Sera was obtained and analyzed by ImmunoCAP for total IgE and specific IgE to α-Gal, beef, pork, cat epithelium and dander, Fel d 1, dog dander, and milk. RESULTS: Forty-five pediatric patients were identified who had both clinical histories supporting delayed anaphylaxis or urticaria to mammalian meat and IgE antibody specific for α-Gal. In addition, most of these cases had a history of tick bites within the past year, which itched and persisted. CONCLUSIONS: A novel form of anaphylaxis and urticaria that occurs 3 to 6 hours after eating mammalian meat is not uncommon among children in our area. Identification of these cases may not be straightforward and diagnosis is best confirmed by specific testing, which should certainly be considered for children living in the area where the Lone Star tick is common.